

Priyanjali Patel

Data Scientist | PhD in Astrophysics

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PhD-trained Data Scientist with hands-on experience in Python-based AI development, specializing in NLP, LLMs, and Generative AI applications. Skilled in building end-to-end AI workflows, including RAG pipelines, chatbots, and automation systems, with experience in LangChain and API integrations. Proficient in developing scalable backend services using FastAPI/Flask, and creating intelligent data pipelines integrating databases and third-party APIs. Experienced in deploying applications using Docker and working with AWS (S3, EC2, SageMaker, CI/CD). Strong in designing automation-driven solutions and translating business needs into production-ready AI systems, with a focus on scalability, efficiency, and real-world impact.

WORK EXPERIENCE

MEDITECH INNOVATIONS • NOIDA WEST, INDIA (Remote) • 08/2025 - Present

Data Science Intern (Big Data & Analytics) – Meditech • Internship

- Built and evaluated **ML/DL models (Logistic Regression, tree-based, DNN, LSTM)** using cross-validation, demonstrating end-to-end model development and performance evaluation.
- Developed NLP systems using LLMs/SLMs, including Hybrid RAG pipelines with vector databases and caching (Redis)
- Applied **hyperparameter tuning (GridSearchCV, RandomizedSearchCV, KerasTuner)** to improve model performance and generalization.
- Developed **end-to-end NLP and time-series pipelines** in Python and PyTorch, covering **data preprocessing, feature engineering, training, and evaluation.**
- Implemented **reproducible and scalable workflows using Docker**, ensuring consistency across development environments.
- **Built and deployed data-driven applications using Streamlit and Flask, with hands-on experience in AWS (SageMaker, S3, EC2, CodePipeline) for scalable workflows and deployment.**
- Tracked experiments and model performance to support **reproducibility, debugging, and iterative improvement.**

Personal Project • 01/2026 - Present

Secure Public Safety Chatbot Platform (FastAPI, JWT, Role-Based Access)

SKILLS

Core Competencies

- Problem Solving
- Cross-functional Collaboration
- Agile Working
- Data-driven Decision Making
- Leadership

Data Science & Machine Learning

- Machine Learning (Supervised/ Unsupervised)
- Statistical Analysis & Modelling
- Feature Engineering & Model Evaluation
- NLP & LLMs (LangChain)
- RAG

Big Data & Data Engineering

- PySpark (Large-scale data processing)
- SQL (Intermediate)
- Data Pipelines & ETL Workflows
- Data Cleaning & Validation
- Distributed Data Processing
- Time-Series Analysis

Programming & Tools

- Python (NumPy, Pandas, SciPy, Astropy)
- Jupyter notebook
- Visual Studio Code
- C
- Github
- C++

- Built a role-based chatbot using **GROQ API KEY**, with JWT authentication and strict access control.
- Secured credentials using bcrypt and backend authorization checks.
- Enabled scenario-based workflows and controlled information sharing for public and authorities.
- Implemented audit logging and access monitoring to reduce security risks.

Collaborative Project • 11/2025 - 01/2026

Machine Learning-based Fraud Detection

- Developed a **binary classification model** for fraud detection using **XGBoost** on an imbalanced dataset.
- Performed **hyperparameter tuning** (learning rate, tree depth, subsampling) to improve model generalization.
- Evaluated model performance using **precision-recall AUC (AUC-PR)** and classification metrics relevant for fraud use cases.

Omdena (Remote) • Palo Alto, United States (Remote) • 09/2023 - 08/2025

Machine Learning Engineer & Collaborator 2025 (3 months) | 2023 (1 month)

Volunteer

- Collaborated in a multi-disciplinary team to build a multi-agent financial chatbot with a modular, agentic design.
- Helped lead the chatbot team, later contributing to the AI team on Insight Engine and Fraud Model integration.
- Used Supervisor agent to orchestrate multi-agent workflows combining RAG + LLM agents and SQL agents using **GROQ API KEY**.

Junior Machine Learning Engineer

Volunteer

- Developed a **U-Net-based deep learning model (PyTorch)** for **glacier segmentation** using multi-band Landsat satellite imagery.
- Processed geospatial data into **256×256 patches with overlap** and generated pixel-wise segmentation masks from polygon labels.
- Addressed **class imbalance** using Dice loss and evaluated performance using Dice coefficient (~0.82).
- Built an end-to-end **semantic segmentation pipeline**, including preprocessing, model training, validation, and visualization.

Data Glacier • 04/2025 - 05/2025

Data Scientist Intern

- Developed classification models (Logistic Regression, Random Forest, Gradient Boosting) to predict client subscription to term deposits.
- Improved minority-class recall using class balancing (class C weights/SMOTE) achieving ROC-AUC of 0.92.

- PyTorch
- scikit-learn

Cloud & Deployment

- Databricks
- AWS (basic exposure – EC2/S3)
- Docker
- API Integration

Data Visualisation and Exploration

- scipy
- matplotlib
- seaborn

Languages

- English
- Hindi
- Spanish

EDUCATION

Ph. D. in Astrophysics

Universidad de Chile, Santiago

Santiago

03/2019 - 12/2024

M. Sc. in Physics

Pt. Ravishankar Shukla

University, Raipur

Raipur

07/2015 - 07/2017

B. Sc. (Physics, Computer Science, Mathematics)

St. Thomas College, Bhilai

Bhilai

06/2012 - 07/2015

- Analyzed confusion matrices & classification reports to optimize precision-recall trade-offs.

Viridien (formerly CGG) + S2DS • London (Remote) • 02/2024 - 03/2024

Science to Data Science S2DS Fellowship (Concurrent with PhD)

- Collaborated in an Agile team of five data scientists on a project for CGG, a global leader in technology and HPC services.
- Designed and developed an LLM-powered GitHub refactoring assistant to streamline developer workflows and save engineering time using **OpenAI API**.

Universidad de Chile, Santiago, Chile • 03/2019 - 12/2024

Doctoral Researcher at Universidad de Chile, Santiago • Full-time

- Analysis of Quasar **Optical Variability using Python and Advanced Statistical Techniques**. Retrieved and processed large-scale time-series data (light curves) for ~7000 quasars **via APIs** from the Zwicky Transient Facility (ZTF) survey.
- Calculated quasar optical variability of using C code (Mexican Hat code) for quasars, **identifying and fixing a critical error to enhance short timescale variability detection**.
- Conducted linear regression analysis with **Orthogonal Distance Regression (ODR)** using the **scipy package** to determine relationships between variables.
- Used median **binning and bootstrapping (1000 samples)** to estimate standard errors and demonstrate anti-correlation between variables.
- Demonstrated that the bending power law model better fits the **stochastic light curve of quasars**, explaining variance dependencies on different parameters.
- Calibrated and processed high-resolution 7 mm **VLBA image data for gravitational lens candidates** using **rPICARD and the CASA** calibration framework.
- Used advanced Bayesian modeling (MCMC) to generate data-driven insights and effectively communicate findings to diverse technical audiences.

Contributions at ALerCE Collaboration

- Built **custom Python pipelines with a ML team for stochastic light curve analysis**, improving efficiency by 15% (saved ~3 hrs/week).

Teaching Assistant

- Graded astronomy exams for five semesters and developed well-structured exams for one semester.

AWARDS & SCHOLARSHIPS

PhD Fellowship, Chile

ANID (National Research and Development Agency)

PUBLICATIONS

P. Patel et al., Probing the rest-frame wavelength dependence of quasar variability, 2025, A&A, 695, A162

EDP Sciences

P. Arévalo et al., Optical variability in quasars: scalings with black hole mass and Eddington ratio depend on the observed time-scales, 2023, MNRAS, 526, 6078

Oxford University Press (OUP)

P. Arévalo et al., The universal power spectrum of quasars in optical wavelengths, 2024, A&A, 684, A133

EDP Sciences